

Test 3 Review

Disclaimer Making up integration problems can be difficult since it is often deceptively difficult to integrate simple surfaces. So I highly recommend using in-class examples (especially ones worked in the video recordings) along with this review to aid in studying.

1 Prior to integrating

Problem 1. Rewrite the following expressions and equations in polar coordinates and graph the first two.

1. $2x^2 + 2y^2 = 19$

2. $x^2 + y^2 = 4x + 6y$

3. $\frac{\sqrt{x^2+y^2}}{xy}$

Problem 2. Rewrite the following in spherical coordinates and graph it.

$$x^2 + y^2 + z^2 = 10y$$

Problem 3. Estimate the volume of the surface $f(x, y) = xy - (x + y)^2$ bounded by the region $R = \{(x, y) : 1 \leq x \leq 5, 0 \leq y \leq 3\}$ with $n = 2$, $m = 2$ (n across the x -axis, m across the y -axis) and

1. Lower left corner

2. midpoint

3. upper right corner

2 Multiple Integrals Set-up

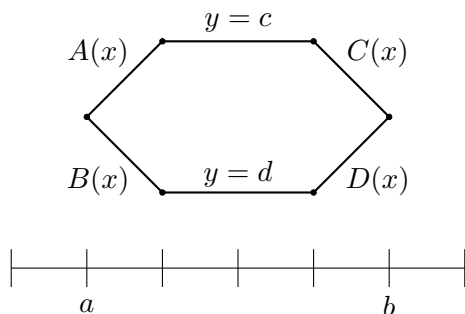
For each of the following problems, set up *but do not evaluate* an expression using integrals which solves the given problem.

Problem 4. The volume under the surface $f(x, y) = e^{\sin xy}$ bounded by $y = 3x - 3$ and $y = x^2 - 3x + 2$

Problem 5. The volume of under the surface $f(x, y) = \frac{x}{y}$ bounded by the region $x^2 + y^2 = 4x$ and $y = x$.

Problem 6. The volume between $z = x^2y^3$ and $z = 1$ and above the region $x^2 + y^2 \leq 7$

Problem 7. The area of the following hexagon.



3 Multiple Integral Evaluate

For each problem, evaluate the multiple integral.

Problem 8. $\int_0^1 \int_0^3 ye^{xy} dx dy$

Problem 9. Evaluate $\iint_R \frac{1}{xy} dA$ where $R = \{(x, y) : 1 \leq y \leq e, y \leq x \leq y^2\}$

Problem 10. Evaluate the following triple integral where E is the solid bounded by the plane $z = 1 - x - y$, above the xy -plane, and between $y = \sqrt{x}$, $y = 0$, and $x = 1$.

Problem 11. Find the volume of the solid which lies between $x^2 + y^2 + z^2 = 25$, above the xy -plane and below $z = \sqrt{x^2 + y^2}$.

Problem 12. Evaluate the following integral where T lies within $x^2 + y^2 = 1$, above $z = 0$ and below $z^2 = 4x^2 + 4y^2$.

$$\iiint_T x^2 dV$$

Problem 13. Evaluate the following integral where R is the region $9x^2 + 4y^2 = 36$.

$$\iint_R x^2 dA$$

Problem 14. Switch the order of integration and evaluate.

$$\int_0^4 \int_{\sqrt{x}}^2 \sin y^3 dy dx$$

4 Extra Things

Problem 15. Find the average value of $\iint_R x dA$ where R is bounded by the first quadrant and $x^2 + y^2 = 4$.

Problem 16. Find the Jacobian of the transformation $x = 2u + 6v$, $y = u^2 + v^2 + w^2$, and $z = uvw$